Rotating target detection algorithm in remote sensing images based on improved YOLOv5s

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***Abstract*—Aiming at the problem of low target detection accuracy and easily missed detection in remote sensing images, a rotating target detection method based on YOLOv5s is proposed with YOLOv5s as the basic framework. First of all, the circular smooth label CSL (Circular Smooth Label, CSL) is used to realize the conversion of angle regression prediction to classification prediction to overcome the problems of regression prediction angle periodicity and variable boundary and improve the detection accuracy. Secondly, the densely coded label (Densely Coded Label, DCL) replaces the sparsely coded label, dramatically reducing the thickness of the prediction layer and improving the training speed. Finally, in the C3 module of the original YOLOv5s feature extraction and feature fusion network, a lightweight channel attention mechanism is introduced into the module, thereby improving the network's local feature capture and fusion capabilities. The experimental results show that on the public DOTA remote sensing image dataset, the map results of the improved YOLOv5s_obb reach 0.78, respectively. Compared with the original YOLOv5s, this method can be improved by 1.7%, and compared with other typical remote sensing target detection methods, the accuracy has also been improved, which proves the effectiveness of the improved YOLOv5s_obb method.**

Keywords-component; remote sensing image; rotating target detection; circular smooth label; attention mechanism

I. INTRODUCTION

Remote sensing images are images obtained by looking down from the sky. In recent years, with the increasing maturity of aerial imaging technology, how to effectively process and apply these images has become a hot topic[1,2]. Target detection is an important research direction in computer vision and has been widely used in various scenarios. However, target detection in current RGB remote sensing images still faces many challenges. In addition, since the satellite image is taken from a high angle, the aspect ratio of the ground detection object is often huge, and the object has a specific directionality. However, if the horizontal box object detector does not consider angle information, it will ignore the directional characteristics of the object. At the same time, the target selected by the horizontal frame also contains a large amount of background information, which is also unfavorable for the accurate positioning of the target.

At present, the target detection algorithm in deep learning can be divided into one stage and two stages: the first stage target detection algorithm is represented by SSD[3], YOLO[4,5,6,7], etc., and the detection speed is fast, but the accuracy is not high. Although the two-stage target detection algorithm represented by the RCNN[8,9,10] series has a high accuracy rate, it needs to be processed twice, increasing time complexity and decreasing detection speed. In addition, Gong[11] introduced a self-attention mechanism in YOLOv5, which achieved better detection results while increasing the computational complexity. Yang[12] et al. introduced a multi-dimensional supervised tension network to remote sensing images to improve the model's ability to detect small targets in complex backgrounds; at the same time, they improved the loss function and proposed IoU-SmoothL1 loss to improve the angle regression results.

The object detection algorithm in the YOLO network model has a high detection speed and can improve the detection accuracy as the number of detection layers of the version increases. The latest YOLOv5 detection model has achieved good results in multiple scene detection tasks[13]. It also encapsulates four model frameworks of different sizes, namely YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x, to meet the needs of different scene detection tasks. Because of the characteristics of the remote sensing image itself and the insufficiency of the general target detection model in its application, this project will comprehensively consider the calculation amount and detection accuracy of the model and, based on the YOLOv5s model framework, research a remote sensing image rotation target detection method based on YOLOv5s.

The YOLOv5 model is a standard natural scene image detection method. However, unlike it, remote sensing images are primarily from a top-down perspective, with denser target distribution, more significant aspect ratio, and more obvious directionality. For targets with a large aspect ratio and a certain angle, the marked background area may be larger than the actual area. The large amount of irrelevant background

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semantic information in the remote sensing image must be fully considered, causing the model to focus too much attention during detection On the useless background instead of the target to be detected. At the same time, due to the large proportion of small objects in remote sensing images, the model does not fully use the feature layer containing semantic information about small objects. According to the problems existing in the detection of remote sensing images by YOLOv5s described above, this paper will focus on the complex background information of remote sensing images; the detection target has a large aspect ratio, has a certain angle and a large number of small targets, and improves the YOLOv5s model. First, for detection targets with a large aspect ratio, a certain angle, and dense targets, CSL is used to solve the problems of Exchangeability of Edges (EoE) and periodicity of angles Periodicity of Angular(PoA). Angle prediction is transformed from a regression problem to a classification problem to improve detection accuracy. Secondly, because of the poor detection effect of small targets caused by a large number of small targets and dense distribution, the channel attention mechanism is introduced to strengthen the local and salient features in the image; based on the above, we have constructed an improved YOLOv5s remote sensing image target detection new method. The main work can be summarized as follows: (1) For the problems of arbitrary and different lengths of detection targets in remote sensing images, redundant background information but less target information to be detected, a large number of small targets, and poor detection effect of the model on small targets, etc. The YOLOv5s method is improved, and a new method for end-to-end learning of remote sensing image target detection is proposed. (2) In the specific implementation of the method, the circular smoothing label (CSL) is used in the YOLOv5s model to classify the angle; in C3, A lightweight incentive squeeze attention module is embedded in the module to enhance the model's attention to the target area. (3) The proposed improved YOLOv5s method is evaluated on the DOTA dataset, and its mAP detection accuracy can be increased by 6.39% compared with the original YOLOv5s respectively, and compared with other mainstream remote sensing target detection methods, the accuracy is also significantly improved.

II. Algorithm Description

A. YOLOv5 Object Detection Algorithm

The whole network structure of YOLOv5 consists of four parts, namely the input end, backbone network, Neck, and prediction end, as shown in Figure 1. At the input end, adaptive scaling can be used to adjust the size of the input image. The backbone network includes Focus, Convolution Match Normalization Lead ReLU (CBL), Bottleneck CSP, and SPP[14], which extract features from images. The Focus model is similar to Downsampling, which concentrates the information on the width and height of one channel while maintaining the feature information of the image. The Bottleneck CSP structure incorporates the concept of residual structure into the model. It combines it with the other two branches to achieve feature fusion at multiple levels, significantly improving the feature extraction performance of the network. The SPP module adopts different pooling structures and multiple perception field fusion technology, improving the model's ability to detect targets in complex scenes. The Neck section uses PAN to construct and sample it up and down to fuse it at the same scale, thereby obtaining more feature information. Three feature vectors of different scales will be obtained at the prediction end to predict the final result.

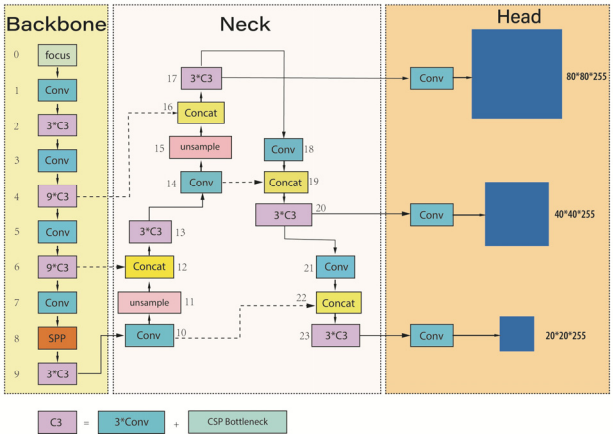


Figure 1. YOLOv5 network structure.

Although YOLOv5 has good detection performance for natural scenes, its application in optical remote sensing images still has the following problems. Firstly, compared to natural scenes, the targets in remote sensing images are mostly high aspect ratio and directional targets. If horizontal boxes are used, the directional information of the targets will be ignored, and it is not conducive to accurate target positioning. Secondly, due to the lack of a practical attention mechanism module in YOLOv5, its detection accuracy for complex backgrounds is low.

B. Introducing CSL

This paper uses CSL[15] to transform the original angle regression prediction problem into a classification problem. In order to describe the directional information of the target, different from the horizontal frame target detection algorithm, the label frame of the rotating frame target detection has additional angle information θ to indicate the directionality. The long side definition method represents a rotating quadrilateral whose center coordinates are (x, y), the long side is $L_{longside}$, and the short is $L_{shortside}$. The label format is (x, y, $L_{longside}$, $L_{shortside}$, θ), where θ is x. The axis is rotated to the angle of the side where the long side is located. The angle range is [-90°, 90°). Clockwise rotation is positive, and counterclockwise rotation is negative, as shown in Figure 2.

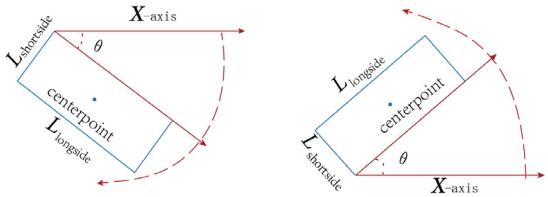


Figure 2. Long edge definition method.

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C. *Embed Stimulus Squeeze Attention Module*

The attention mechanism has been widely used in target detection in recent years. Adding the attention mechanism to the detection network can reduce the influence of complex background information on the detection network, making the detection network more focused on detecting the target information, thereby improving detection efficiency. Performance. YOLOv5s is a general-purpose target detection model that uses natural images as objects. When processing remote sensing image detection tasks, it must fully consider the characteristics of extensive backgrounds and small targets in remote sensing images. As a result, the feature extraction of the model network cannot fully pay attention to small target areas. There is a phenomenon of minor target errors and omissions. This project intends to embed the Stimulus Squeeze Attention (SE) module[16] in the YOLOv5s backbone network architecture to make the model pay more attention to the location and category of the object to be detected and reduce redundant background information.

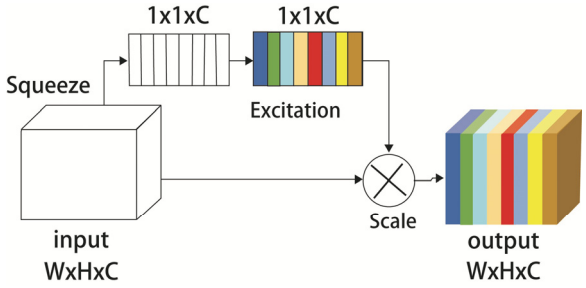


Figure 3. SE attention module structure.

The SE module includes two operations, one is Squeeze, and the other is Excitation. In Figure 3. C represents the channel number of the input characteristic map, W represents the width of the characteristic map, and H represents the height of the characteristic map; that is, the size of the input characteristic map is W×H×C. First, using the global average pooling (GAP) technique, the remote sensing image feature map is compressed into a 1×1×3 channel vector, and the channel weights are extracted. The excitation operation consists of two layers of fully connected layers. The first layer is excited by C × Ratio neurons to obtain 1 × 1 × C × Radio output (Radio is a scaling parameter, and the computational complexity can be reduced by controlling the value of Radio). The second fully connected layer will take the output of the first fully connected layer as its input, and the second layer will be excited by C neurons to get 1×1×C×Radio output. On this basis, the vector is combined with the original feature map, and the weight of each channel calculated by the SE module is multiplied by the corresponding channel of the original feature map to obtain important information about the image. In this paper, the SE module is embedded into the CSP Bottleneck in Backbone so that the model can better extract the features of the remote sensing image detection target. The module is named the C3_SE module. Figure 4. is a schematic diagram of the structure of the SE module embedded in the CSP Bottleneck.

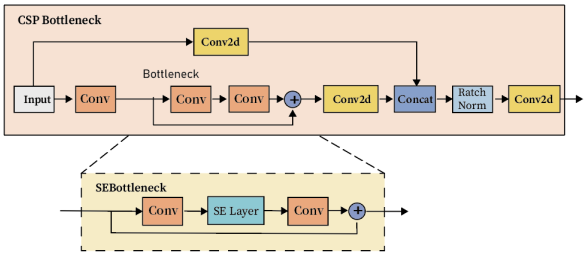


Figure 4. Diagram of embedding attention module.

D. *Improved angle regression*

This algorithm transforms the target rotation detection from a regression problem to a classification problem[17], thus solving the problems of EoE and PoA in regression prediction. However, the CSL method also has apparent defects; that is, the prediction head of 180 angle classes needs to be more bulky, resulting in too many training parameters, seriously affecting the rotation detector's training effect. This project plans to adopt the DCL method based on Gray code to solve this problem. Compared with the CSL method based on sparse coding, the binary codes of adjacent two digits in Gray code are only one digit apart, significantly reducing the training time and improving detection accuracy.

The dense coding based on Gray code can effectively shorten the coding length and has a wide range of representations to solve the problem of a too-thick prediction layer caused by a too-long CSL coding length and improve the training efficiency. Based on the regression method, CSL method, and Gray code, the calculation formula of the dense coding prediction layer thickness is deduced as follows:

$$Th_{reg} = A. \tag{1}$$

$$Th_{csl} = A \times AR/\omega. \tag{2}$$

$$Th_{gcl} = A \times [\log_2(AR/\omega)]. \tag{3}$$

In the formula, A is the number of anchor boxes, AR (Angle Range) represents the range of angles, $\omega = AR/C_\theta$ represents the degree of discretization of angles, and C_θ represents the number of angle categories. Taking A=21, AR=180, and ω=1 as an example, the regression obtained 21 layers, CSL obtained 3780 layers, which is 180 times the regression prediction, and the Gray code obtained 168 layers, which is only eight higher than the regression prediction times, proving that dense coding can significantly reduce the predicted layer thickness.

III. Experiment

A. *Dataset Introduction and Preprocessing*

The DOTA1.0 dataset[18] has 2806 high-resolution images ranging from 800x800 to 20000x20000, including 15 objects and 188282 samples. Because the pixels of the picture are relatively large, if it is directly input into the model for training, the compression will reduce the clarity of the image and the detection accuracy, so the picture is cut to a size of 1024×1024.

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The GAP is set to 50%. The number of cropped images increases from 2806 to 39589.

B. Experimental Environment And Training Strategy

The experiment is carried out in the Ubuntu17.04.5 operating system. The hardware configuration is a 12-core CPU Intel(R) Xeon® Platinum 8255C. GPU is Nvidia RTX 2080Ti 11G graphics card. The software version is Pytorch1.9.1, CUDA11.4, and Python3.8. The experiment adopts the stochastic gradient descent (SGD) optimizer; the momentum and weight decay are set to 0.937 and 0.0005, respectively, the batch size is 16, and the iterative training is 300 generations; the learning rate is set to 0.001 in the first three rounds and then reaches 0.01, and Continue to decay to 0.0001 until the end of the training.

C. Experimental Results and Analysis

In order to verify the effect of the proposed method on remote sensing image target detection, the ablation experiment of the improved module was first carried out on the DOTA1.0 dataset, and the experimental results are shown in TABLE I.

TABLE I. ABLATION EXPERIMENT RESULTS MAP

Method	SE	CSL	DCL	DOTA	Parameter（m）	training (h)
YOLOv5s				70.23	7.2	22
YOLOv5s_A	√			70.53	7.42	23
YOLOv5s_B		√		76.10	7.5	52
YOLOv5s_C			√	76.32	7.5	23
ours	√	√	√	76.62	7.72	24

In Table I, YOLOv5s_A, YOLOv5s_B, and YOLOv5s_C represent adding SE module, introducing CSL, and improving angle regression respectively.

1) When the SE attention module is added alone, the mAP is increased by 0.30%, the number of parameters is increased by 3%, and the training time is 23h.

2) The mAP of the model using CSL has increased by 5.87%, the number of parameters has increased by 4.2%, and the accuracy has improved significantly, but the training time has also increased significantly, reaching 52h.

3) Using the DCL model to train mAP increased by 6.09%, the number of parameters increased by 4.2%, and the training time was greatly reduced compared with CSL, only 23h.

It can be seen that the detection accuracy is increased by 0.3% when only the SE attention module is added, and the method of converting angle regression to angle classification using CSL is also feasible, and its average accuracy is increased by 5.87%, which is remarkable. But the training time has increased a lot, which is not friendly to users with low GPU performance. On this basis, using the DCL algorithm to improve, not only the accuracy rate increase by 6.39% but also the time spent is almost the same as regression training.

The comparison results of the algorithm in this paper and other algorithms are shown in TABLE II. In the target detection papers of the DOTA1.0 data set in recent years, a single-stage detection algorithm and a two-stage detection algorithm were selected for comparison, and it was found that the average accuracy of the algorithm is higher than other algorithms.

TABLE II. COMPARISONS WITH OTHER WORKS

Model	One-stage	Two-stage	Map/%
Mask R-CNN [19]		√	71.6
Faster R-CNN **[10]**		√	70.8
APE [20]		√	75.75
FPN-CSL [21]		√	76.17
RetinaNet [22]	√		67.5
DRN [23]	√		73.25
R^3Det [24]	√		73.74
RSDet [25]	√		74.2
ours	√		76.62

The visual display of the experimental results is shown in Figure 5. The algorithm in this paper has a very good prediction effect on the angle, the angle classification is accurate, and the dense small target detection is relatively complete, which can basically meet the target detection of remote sensing images.



Figure 5. Visualization results of the algorithm in this paper on the DOTA dataset.

IV. CONCLUSIONS

Given the complex background information of remote sensing images, many tiny targets, and the dense arrangement, a remote sensing image detection method based on YOLOv5s is proposed. First, embedding an incentive-squeezing attention module in the model makes the model pay more attention to the location and category of the detected object, thereby reducing

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redundant background information. Secondly, add angle prediction and change the regression method prediction to classification method prediction; finally, use DCL to improve the training speed. On the DOTA1.0 data set, compared with the original YOLOv5s model, the average accuracy rate has increased by 6.39%. Using After DCL, the training speed remains the same. In future work, consider introducing a Transformer with stronger feature extraction capabilities module to enhance the model's perception of global semantic information, and on the other hand, it will try to explore a new prediction frame loss function to improve the positioning effect of the model prediction frame.

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